

CCSDS 235.1-W-0.4 Reference Implementation Competition

Timeline, Deliverables, Gateway Requirements & Scoring

Sponsored by ESA — Organised by AMSAT-UK

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1 Competition Key Dates

Date	Event	Notes
5 April 2026	Registration and Team Formation Closes	Final deadline for new registrations
Late June / Early July 2026	Workshop at Goonhilly Earth Station	Collaborative session, interoperability testing
31 August 2026	Competition Closes	Final submission deadline; judging begins
30 September 2026	Judging complete and results published	Winners Announced
October 2026	CCSDS Meetings	Winners attend CCSDS meetings

The competition runs for approximately 21 weeks from registration close to final submission.

Attendance at the Goonhilly workshop is not mandatory but is strongly encouraged. Teams attending must bring working implementations of Gateways 1–3 (SPDU Layer, COP-P Layer, Frame Layer) and the ability to produce and consume hex dumps for cross-team wire format exchange.

2 Development Timeline

The schedule is structured around the Goonhilly workshop as the mid-point milestone. Gateways 1–3 receive the most development time because they form the protocol core and are prerequisites for workshop participation.

Important: The dates shown for each gateway are suggested timings to help teams plan their development. They are not marking points or deadlines — teams will not be assessed on whether they hit individual gateway dates. Only the final submission deadline (31 August 2026) is a hard deadline.

2.1 Phase 1: Foundation and Core Protocol (Weeks 1–10)

Gateway	Capability	Dates	Duration
0	Design and Architecture	5 Apr – 19 Apr	2 weeks
1	SPDU Layer	20 Apr – 10 May	3 weeks
2	COP-P Layer (FOP-P, FARM-P)	11 May – 31 May	3 weeks
3	Frame Layer	1 Jun – 14 Jun	2 weeks

2.2 Phase 2: Workshop Preparation (Weeks 11–12)

Activity	Dates	Duration
Workshop preparation and buffer	15 Jun – 28 Jun	2 weeks
Generate 5 interoperability test artifacts	Within buffer	–
Practice hex dump exchange and live demos	Within buffer	–

2.3 Workshop at Goonhilly Earth Station (Late June / Early July)

Cross-team wire format exchange, COP-P behaviour verification, live protocol exchange, standards clarification sessions.

2.4 Phase 3: Session Control and Integration (Weeks 13–18)

Gateway	Capability	Dates	Duration
4	Physical Layer Abstraction	4 Jul – 12 Jul	1.5 weeks
5	State Machine	13 Jul – 26 Jul	2 weeks
6	Hailing and Session Control	27 Jul – 9 Aug	2 weeks

2.5 Phase 4: Validation and Documentation (Weeks 19–21)

Gateway	Capability	Dates	Duration
7	Integration Testing	10 Aug – 16 Aug	1 week
8	Conformance Testing	17 Aug – 21 Aug	1 week
9	Interoperability Testing	22 Aug – 25 Aug	0.5 weeks

Gateway	Capability	Dates	Duration
10	Documentation and Examples	26 Aug – 31 Aug	1 week

Note: Integration testing, conformance testing, and documentation should be developed incrementally throughout the competition, not only during the final phase. The dates above represent the dedicated focus periods for finalising each gateway.

3 Deliverables and Gateway Requirements

The dates shown in each gateway heading below are suggested timings, not marking points. Teams are free to work at their own pace provided all deliverables are submitted by 31 August 2026.

3.1 Gateway 0: Design and Architecture (5 Apr – 19 Apr)

Deliverables: System architecture document, technology stack selection, development environment setup, testing strategy.

Validation: Architecture is documented with defined components and layer boundaries. Build system compiles and test framework runs.

3.2 Gateway 1: SPDU Layer (20 Apr – 10 May) — Section 3

Deliverables:

- SPDU encoder for all types (F1, F2, 1–5)
- SPDU decoder for all types
- PLCW generator (Type F1 and F2)
- Directive decoder (Type 1 and 2)
- Big-endian byte order handling
- Unit test suite

Validation: All SPDU types encode and decode correctly. PLCWs match specification format. Malformed SPDUs rejected gracefully. Encoding and decoding completes in under 1 ms per SPDU.

3.3 Gateway 2: COP-P Layer (11 May – 31 May) — Section 6, Section 4

Deliverables:

- FOP-P implementation per Section 6.2 (8-bit and 16-bit sequence numbers)
- FARM-P implementation per Section 6.3
- COP-P coordinator (Expedited and Sequence Controlled services)
- Persistence activity process per Section 4
- Unit test suite

Validation: Sequence numbers maintained and wrap correctly. Retransmission recovers lost frames. Window management prevents overflow. Resynchronization resets state correctly (SET V(R) per Section 6.2.3.2).

3.4 Gateway 3: Frame Layer (1 Jun – 14 Jun) — Section 5.4–5.6

Deliverables:

- Frame transmitter (P-frames and U-frames)
- Frame receiver and parser
- QoS support (Expedited, Sequence Controlled)
- Version-3 and Version-4 frame support
- Unit test suite

Validation: P-frames and U-frames distinguished correctly. SPDUs extracted from P-frames. QoS flags set per specification. Frame validation detects malformed frames. Frame processing under 1 ms per frame.

3.5 Gateway 4: Physical Layer Abstraction (4 Jul – 12 Jul) — Section 5.4

Deliverables:

- Abstract physical channel interface with PCID support
- Mock implementation (in-memory loopback for unit testing)
- Network-based implementation (TCP or UDP for ground testing)
- Channel status reporting
- Unit test suite

Validation: Mock channel enables deterministic testing. Network channel enables multi-host interoperability. PCID correctly identifies channels.

3.6 Gateway 5: State Machine (13 Jul – 26 Jul) — Section 5.1–5.3

Deliverables:

- State machine implementation per Section 5
- State tables for Full Duplex (Tables 5-1, 5-2), Half Duplex (Tables 5-1, 5-3), Simplex (Tables 5-1, 5-4)
- State transition validation
- State history logging
- Unit test suite

Validation: All state transitions match CCSDS 235.1 tables exactly. Invalid transitions rejected. All three duplex modes operate per specification.

3.7 Gateway 6: Hailing and Session Control (27 Jul – 9 Aug) — Section 5.1.3, Section 4

Deliverables:

- Hailing mechanism (caller and responder roles) per Tables 5-7, 5-10
- Caller and responder controller implementations
- Session lifecycle management
- Reconnect (rehailing) per Tables 5-9, 5-12
- COMM_CHANGE support per Tables 5-8, 5-11
- Unit test suite

Validation: Hailing establishes link per specification. Caller and responder roles operate correctly. Reconnect recovers from communication loss. Session lifecycle completes per specification.

3.8 Gateway 7: Integration Testing (10 Aug – 16 Aug)

Deliverables:

- Integration test suite (complete session lifecycle, all duplex modes, reconnect, resynchronization, multi-channel)
- Performance benchmarks (throughput, latency, memory)

- Stress tests (high load, multiple sessions, long duration)

Validation: All integration tests pass. Performance meets targets (under 1 ms per frame, under 50 KB per session). No memory or resource leaks.

3.9 Gateway 8: Conformance Testing (17 Aug – 21 Aug)

Deliverables:

- Conformance test suite for all mandatory CCSDS 235.1-W-0.4 features
- Protocol Implementation Conformance Statement (PICS)
- Compliance validation report
- Test vectors in standardised formats (binary + JSON metadata)

Validation: All mandatory features implemented. All conformance tests pass. PICS document complete and accurate.

3.10 Gateway 9: Interoperability Testing (22 Aug – 25 Aug)

Deliverables:

- Interoperability test suite (SPDU exchange, session establishment, data transfer, error recovery)
- Test vector exchange capability in standardised format
- Interoperability report

Validation: Successful session with other implementations. Test vectors decode correctly across implementations.

3.11 Gateway 10: Documentation and Examples (26 Aug – 31 Aug)

Deliverables:

- API documentation (all public interfaces)
- User guide (build, configuration, usage, troubleshooting)
- Architecture documentation (design decisions, component interactions, data flow)
- Working example programs (caller station, responder station)
- Wire format specification

Validation: Build instructions enable anyone to build and run the implementation. All public APIs documented. Examples demonstrate key functionality.

3.12 Workshop Interoperability Artifacts

Teams attending the Goonhilly workshop must produce these 5 artifacts as hex dumps:

#	Artifact	Specification
1	Type F1 PLCW	Report Value=127, Retransmit=false, PCID=0, Expedited Counter=3
2	Type F2 PLCW	Report Value=500, Retransmit=true, PCID=1, Expedited Counter=6

#	Artifact	Specification
3	Variable-Length SPDU	Type 1 Directive, SET V(R) with SEQ_CTRL_FSN=42
4	P-frame	Version-3, containing artifact #1, Expedited QoS
5	U-frame	Version-3, 10-byte payload 0x00–0x09, Sequence Controlled QoS, seq=7

4 Submission Requirements

4.1 What to Submit

Required:

1. Complete source code in your chosen language
2. Build system and build instructions
3. Test suite with results
4. Documentation (API, user guide, architecture)
5. Examples demonstrating functionality
6. Conformance validation report
7. PICS document

Optional but Recommended:

- Performance analysis report
- Interoperability test results
- Additional test vectors
- Visualisation tools (state machine, session lifecycle, etc.)

4.2 Validation Data to Include

From Specification:

- CCSDS 235.1-W-0.4 specification document
- Test vectors you generated

Your Results:

- SPDU encoding/decoding test results
- COP-P test results (FOP-P, FARM-P)
- Session lifecycle test results
- Conformance test results
- Performance benchmark results
- Interoperability test results

4.3 Format Requirements

- Clear directory structure
- README at root with overview
- Reproducible build process
- All dependencies documented
- Version control (Git)

4.4 Flexibility

You have complete freedom in:

- Programming language choice
- Implementation details and algorithms
- Internal architecture and design
- Tools and libraries

- File organisation
- Optimisation strategies

You must comply with:

- CCSDS 235.1-W-0.4 specification (all mandatory features)
- Output correctness (SPDUs, frames, state transitions)
- Performance targets (under 1 ms per frame, under 50 KB per session)
- Deliverable completeness (all gateways)

4.5 How Submissions Are Validated

Your implementation will be validated by:

- Running your test suite and checking results
- Testing with known inputs and comparing outputs
- Measuring performance (throughput, latency, memory)
- Checking CCSDS 235.1 conformance
- Interoperability testing with other implementations

5 Scoring Scheme

Submissions are scored out of a maximum of 100 points across five categories.

5.1 Category 1: Protocol Correctness (40 points)

The largest weighting reflects the primary goal of the competition: a correct, specification-compliant implementation.

Criterion	Points	Description
SPDU encoding/decoding (Gateway 1)	8	All SPDU types (F1, F2, 1–5) encode and decode correctly per Section 3. Big-endian byte order. Malformed SPDU handling.
COP-P protocol (Gateway 2)	10	FOP-P and FARM-P operate per Section 6. Sequence numbering, window management, retransmission, resynchronization, persistence. Both 8-bit and 16-bit sequence numbers.
Frame layer (Gateway 3)	6	P-frame and U-frame construction per Section 5. QoS flags, Version-3/4 support, CRC validation.
State machine (Gateway 5)	8	All state transitions match CCSDS 235.1 tables exactly. Full, Half, and Simplex duplex modes. Invalid transitions rejected.
Session control (Gateway 6)	8	Hailing, caller/responder roles, reconnect, COMM_CHANGE per Section 5.1.3. Complete session lifecycle.

5.2 Category 2: Testing and Conformance (25 points)

Criterion	Points	Description
Unit test coverage	8	Comprehensive unit tests for all protocol layers. Target: greater than 90% coverage.
Integration tests (Gateway 7)	5	End-to-end session lifecycle, all duplex modes, reconnect, resynchronization, multi-channel.
Conformance tests (Gateway 8)	7	All mandatory CCSDS 235.1-W-0.4 features tested. PICS document complete and accurate.
Test vectors (Gateway 8)	5	Standardised test vectors generated. Binary + JSON metadata format. Vectors validate correctly.

5.3 Category 3: Interoperability (15 points)

Criterion	Points	Description
Workshop artifacts	5	All 5 interoperability artifacts produced correctly. Hex dumps byte-identical with specification.
Cross-implementation testing (Gateway 9)	5	Successful session establishment and data exchange with at least one other implementation.
Test vector compatibility	5	Test vectors decode correctly when consumed by other implementations. Interoperability report submitted.

5.4 Category 4: Performance (10 points)

Criterion	Points	Description
Throughput and latency	4	SPDU encoding/decoding under 1 ms. Frame processing under 1 ms.
Resource efficiency	3	Session memory overhead under 50 KB. No memory or resource leaks under stress.
Scalability	3	Support for 10 or more concurrent sessions. Sustained operation under load.

5.5 Category 5: Documentation and Usability (10 points)

Criterion	Points	Description
API documentation (Gateway 10)	3	All public interfaces documented. Clear parameter descriptions and usage examples.
User guide and build instructions	3	A reviewer can build and run the implementation from the documentation alone.
Architecture documentation	2	Design decisions explained. Layer interactions and data flow documented.
Working examples	2	Caller and responder examples run successfully. Demonstrate key protocol features.

5.6 Scoring Summary

Category	Maximum Points
Protocol Correctness	40
Testing and Conformance	25
Interoperability	15
Performance	10
Documentation and Usability	10
Total	100

5.7 Bonus Points (up to 5)

Bonus points may be awarded at the judges' discretion for:

- Exceptional code quality or elegant design
- Novel approaches to testing or validation
- Contributions to specification clarification during the workshop
- Tools or visualisations that aid understanding of the protocol (e.g. state machine visualiser, session timeline viewer)

5.8 Partial Credit

Partial credit is awarded within each criterion. A team that completes Gateways 0–6 with solid correctness and testing but does not finish Gateways 7–10 will still score well in Protocol Correctness and may earn partial marks in Testing and Documentation for work completed incrementally.

Gateways are evaluated independently — a strong Gateway 1 is not diminished by an incomplete Gateway 9.

6 Validation Checklist

Use this checklist to verify your submission is complete.

6.1 Gateway 1: SPDU Layer

- ☐ Type F1 PLCW encodes/decodes correctly
- ☐ Type F2 PLCW encodes/decodes correctly
- ☐ Variable-length SPDUs (Types 1–5) encode/decode correctly
- ☐ PLCWs generated from FARM-P state
- ☐ Directives decoded and processed
- ☐ Big-endian byte order
- ☐ Unit tests pass

6.2 Gateway 2: COP-P Layer

- ☐ FOP-P maintains V(S) and N(N) correctly
- ☐ FOP-P transmit window works
- ☐ FOP-P processes PLCWs correctly
- ☐ FOP-P retransmits on request
- ☐ FARM-P maintains V(R) correctly
- ☐ FARM-P detects sequence gaps
- ☐ FARM-P sets retransmit flag
- ☐ COP-P Coordinator provides Expedited service
- ☐ COP-P Coordinator provides Sequence Controlled service
- ☐ Resynchronization works
- ☐ Unit tests pass

6.3 Gateway 3: Frame Layer

- ☐ P-frames transmitted correctly
- ☐ U-frames transmitted correctly
- ☐ Frames received and parsed correctly
- ☐ P-frames vs U-frames distinguished
- ☐ SPDUs extracted from P-frames
- ☐ QoS flags set correctly
- ☐ Version-3 and Version-4 frames supported
- ☐ Unit tests pass

6.4 Gateway 4: Physical Layer

- ☐ PhysicalChannel interface defined
- ☐ Mock channel works for testing
- ☐ Network channel enables ground testing
- ☐ PCID support works
- ☐ Channel status reporting works
- ☐ Unit tests pass

6.5 Gateway 5: State Machine

- ☐ State-controlling variables implemented
- ☐ State tables for Full Duplex
- ☐ State tables for Half Duplex
- ☐ State tables for Simplex
- ☐ State transitions validated
- ☐ State history logged
- ☐ Unit tests pass

6.6 Gateway 6: Hailing and Session Control

- ☐ Hailing works (caller and responder)
- ☐ Caller controller works
- ☐ Responder controller works
- ☐ Session lifecycle completes
- ☐ Reconnect works
- ☐ COMM_CHANGE works
- ☐ Unit tests pass

6.7 Gateway 7: Integration Testing

- ☐ Complete session lifecycle tests pass
- ☐ Full Duplex tests pass
- ☐ Half Duplex tests pass
- ☐ Simplex tests pass
- ☐ Reconnect tests pass
- ☐ Resynchronization tests pass
- ☐ Multi-channel tests pass
- ☐ Performance benchmarks complete
- ☐ Stress tests pass

6.8 Gateway 8: Conformance Testing

- ☐ All mandatory features implemented
- ☐ All conformance tests pass
- ☐ PICS document complete
- ☐ Test vectors generated
- ☐ Compliance report shows 100% conformance

6.9 Gateway 9: Interoperability Testing

- ☐ Session established with other implementations
- ☐ Data exchanged successfully
- ☐ Test vectors compatible
- ☐ Error recovery works across implementations
- ☐ Interoperability report complete

6.10 Gateway 10: Documentation and Examples

- ☐ API documentation complete
- ☐ User guide complete
- ☐ Architecture documentation complete
- ☐ Examples work correctly
- ☐ Performance analysis complete

7 Summary

Phase	Gateways	Period	Focus
Foundation	0–3	Apr – mid Jun	Core protocol layers (SPDU, COP-P, Frame)
Workshop	–	Late Jun / Early Jul	Cross-team testing at Goonhilly
Session Control	4–6	Jul – early Aug	Physical layer, state machine, hailing
Validation	7–10	Aug	Integration, conformance, interop, docs
Submission	–	31 August 2026	Final deadline

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